BRAC UNIVERSITY
Department of Computer Science and Engineering

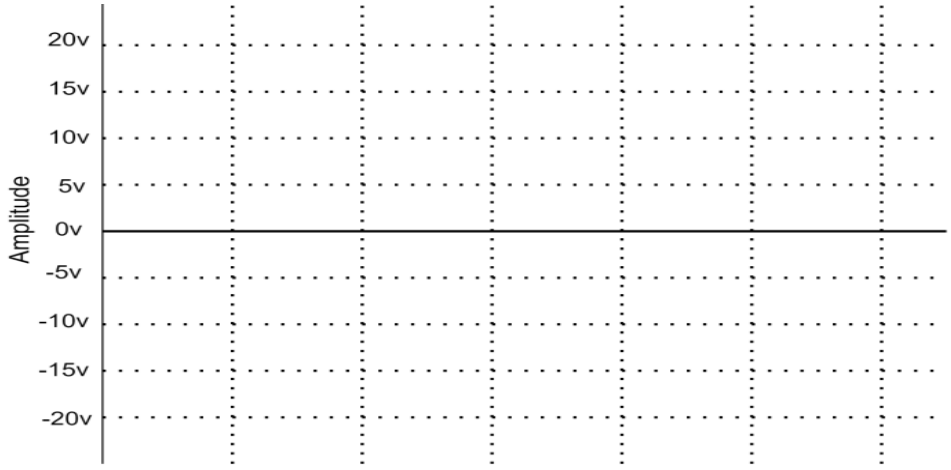
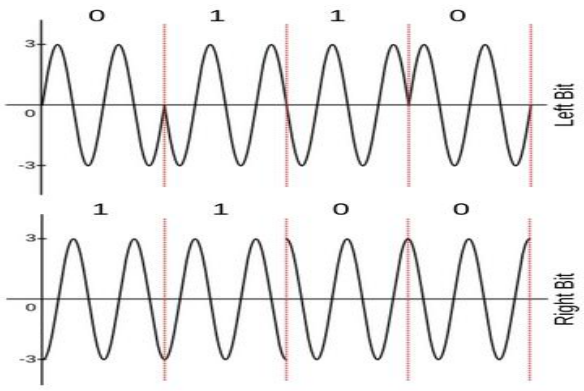
Examination: Semester Final | Semester: Summer 2024
Duration: 2 hours | Full Marks: 50

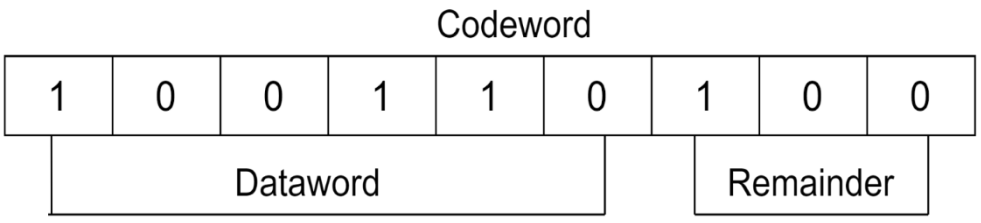
CSE 320/EEE361/ECE361: Data Communications

Answer the following questions.
Figures in the right margin indicate marks.

SET B

Name:	ID:	Section:

1. [CO2]	a)	Consider a multi-level ASK where 2 bits are sent in each baud. The carrier signal has 2 full-cycles in each signal element, and a phase of 0°. Amplitude for each 2-bits pattern is given as: 00 → 5V, 01 → 10V, 10 → 15V, 11 → 20V. Based on this, **draw** the modulated signal **in the following graph** for the bit sequence 111010000100. 	[5]
	b)	For the QPSK-modulated signal given below, **draw** the constellation diagram, and **find** the **amplitude** and **phase** of the modulated signal. 	[2+1+2]

2. [CO3]	a)	The minimum bandwidth required to multiplex 6 channels is 565 kHz. There are 15 kHz guard bands used to prevent interference for each channel. **Calculate** the bandwidth of each input channel. **Show** the required bandwidth with a **diagram** where the starting frequency is 201 kHz. What will be the highest frequency?	[2+2 +1]
	b)	**In which case**, would you prefer using synchronous TDM over statistical TDM though there are empty slots in frame.	[2]
	c)	A video conferencing company adaptively manage bandwidth among multiple clients, ensuring that resources are allocated based on real-time demand rather than fixed slots, leading to enhanced performance and user experience. **Based on the scenario, which TDM is used and why?**	[3]
	d)	Consider 3 channels, each having different input data rates. If the data rate of the first channel is 24 kbps, then the rest of the channels have double the data rate of their previous channel. You need to multiplex them in such a way, so that the number of channels increases. The interleaved unit is 8 bits along with 1 bit for synchronization. **Calculate** the following: i. Draw the figure to visualize the multiplexing process with proper labels ii. Input bit duration iii. Frame Size iv. Frame Rate v. Output bit duration	[5]
3. [CO4]	a)	The following codeword is received by a receiver X. Receiver X uses CRC technology to detect errors. **Determine** whether there is any error in the received codeword or not with proper calculation. [The first and last bits in the divisor are 1, rests are 0] 	[5]
	b)	Suppose we are using a block coding scheme with a (4,6) code. How many Dataword can be **encoded**? How many Codewords will be **invalid**?	[1+1]

	c)	Suppose we are using a (2,4) encoding scheme and the assignment of Dataword to Codewords are as follows: **Dataword** / **Codeword** 00 — 0010 01 — 1001 10 — 1101 11 — 1100 i. What is the minimum Hamming distance here? ii. How many bits can be corrected and detected? iii. The receiver receives 1111. Explain what type of error this is. Can this codeword be corrected? **Justify** your answer.	[2+2+1+3]
4. [CO4]	a)	There are three stations sharing a communication link. Station 1 is on the left side, Station 3 is on the right side, and Station 2 is in between. Assume they are using CSMA/CD. • Station 1 starts sending a message to Station 3 at time **t = 8 ms** • The first bit of the message reaches Station 2 at time **t = 24.8 ms** and Station 3 at time **t = 32.2 ms** • Station 1 finishes sending the message at time **t = 40.8 ms** Based on the given information, **calculate** the followings: i. How long does it take Station 1 to send the message **(Frame Transmission time)?** ii. What is the longest time it takes the message to travel between stations **(Maximum Propagation time)?** iii. If the link speed is **4 Gbps**, what is the **Minimum Frame Size** required to meet the conditions given?	[2+2+3]
	c)	Why is the vulnerable time in pure ALOHA **doubled** of the slotted ALOHA?	[3]

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